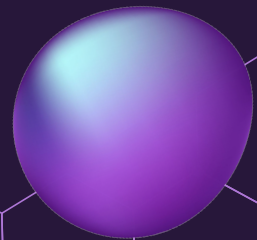
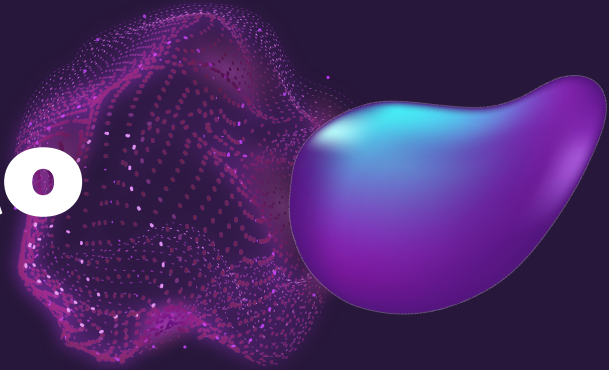
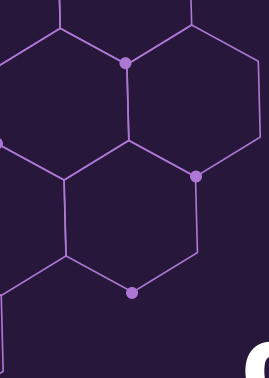


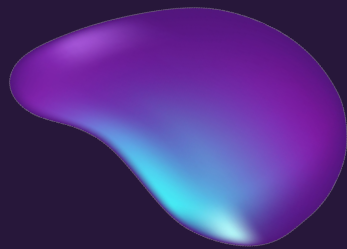
Introduction to Artificial Intelligence

4th & 5th of September 2024





TODAY'S AGENDA



01

ROUND OF PRESENTATIONS

Let's introduce each other. What are your expectations about VU? How is it going so far...?

02

PROJECT GROUPS' MAKING

Form groups with one student per skill, randomly.

03

RESEARCH PROBLEM ASSIGNMENT

Recap on choosing a research gap.

04

GET TO KNOW YOUR GROUP

Socialize, exchange contacts, and brainstorm research problems.

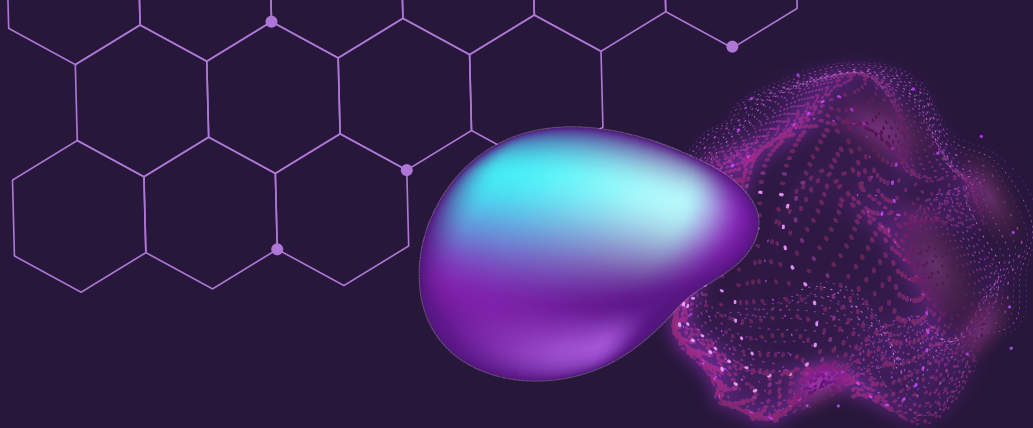
05

FOR NEXT TIME...

5-min presentation on your research problem for feedback.

01

**ROUND
PRESENTATIONS**



OF





Your name, age & origin



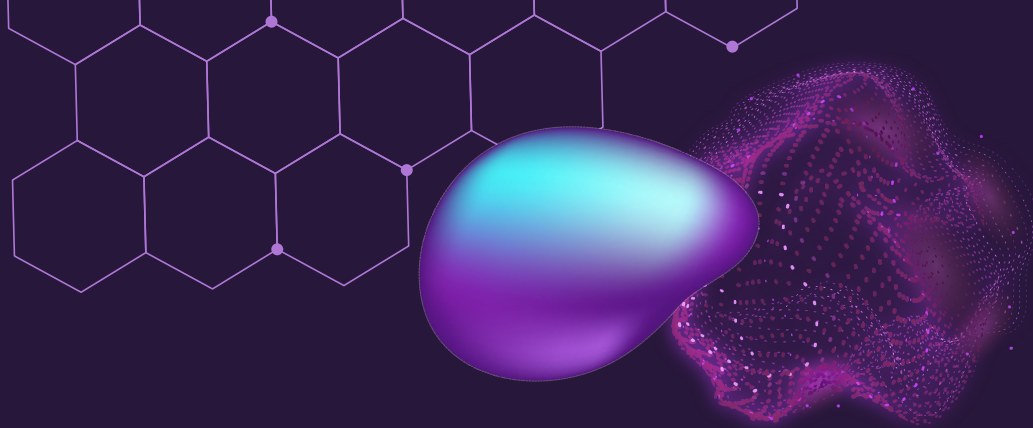
3 (interesting) things about yourself



Your expectations about VU...
Why are you here?

02

**PROJECT
MAKING**

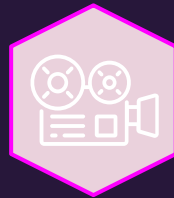


GROUPS'

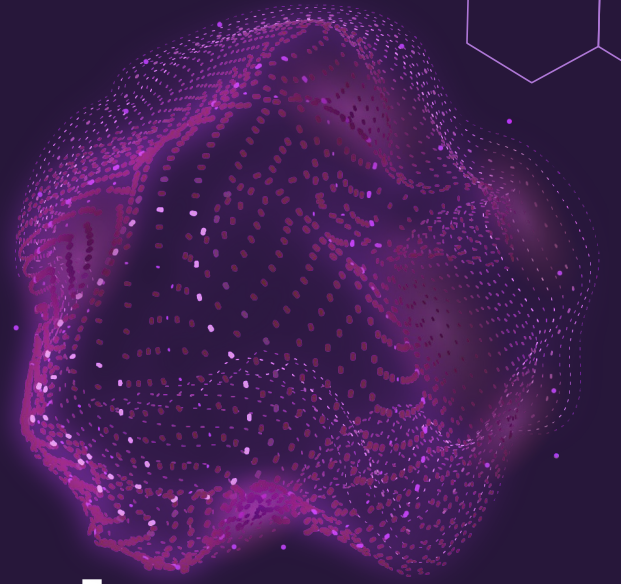


Your Best Skill





5 People 5 Skills 1 Group Project





03

RESEARCH PROBLEM ASSIGNMENT

Identifying a Research Problem

1. Understanding what a Research Problem is:

- **Definition:** A specific issue or challenge within a domain that lacks an adequate solution.
- **Example:** "Detecting depressive behaviors on social media platforms."

**WHAT IS THE DOMAIN?
WHAT IS THE PROBLEM?**

Research Problem (groups assignment)

Published

Assign to

Edit

⋮

Write in 1 A4 what your choice is for the scope of your project (the "research problem") and what the problem is that you want to solve within that domain (the "task").

Examples:

- Support for deafening by converting sound to image,
- Elderly support when using email,
- Detecting depressive people on Facebook
- ...

Tip: Think in your own environment, whether there are people who you could help with an Artificial Intelligence solution.

Take into account: domain, problem, input/output description (of the application), the knowledge that is needed.

NB This is a group assignment, so just one of you will have to submit.

2. Steps to Identify a Problem:

- **a. Choose a Relevant Domain:**
 - Look for an area you're passionate about (e.g., healthcare, education, environment).
- **b. Investigate Current Solutions:**
 - Research existing solutions. Are they effective? What are their limitations?
- **c. Identify the Gaps:**
 - Look for unsolved issues or problems with current solutions (e.g., accessibility, accuracy, scalability).



Depression Detection on Social Media with Large Language Models

SOLUTION

Xiaochong Lan, Yiming Cheng, Li Sheng, Chen Gao, Yong Li
Department of Electronic Engineering, BNRist, Tsinghua University, China lanxc22@mails.tsinghua.edu.cn,
{chgao96, liyong07}@tsinghua.edu.cn

5 Conclusion and Future Work

EFFECTIVENESS?

In this work, we introduce a novel depression detection system named DORIS, which incorporates clinical practices in depression diagnosis, including the widely used diagnostic criteria DSM-5, and analyses of mood course, achieving high accuracy and interpretability in depression detection. Extensive experimental results demonstrate the effectiveness of our method, as well as the rationality of each design component in DORIS.

In the future, we plan to deploy our system in real-world settings to collaborate with human experts in clinical diagnosis. This will enable us to identify areas for improvement in our system and continue its iterative design.

Limitations

SHORTCOMINGS?

- **Our experiment is based on a single dataset.** Large-scale datasets with expert annotations are challenging to acquire in the field of depression detection, and we find only one dataset that meets our criteria, further refined based on our standards. The experimental results on this dataset have demonstrated the effectiveness of our method. However, we believe that experiments on more datasets could further validate our approach.
- **Our focus is solely on depression.** Mental health analysis and mental illness detection are crucial for human well-being. Our work concentrates on only one of these tasks: depression detection. While depression is a major mental illness, extending our method to detect other mental health illnesses, such as anxiety and bipolar disorder, would be more meaningful. Specifically, our explicit modeling of the mood course could be significantly beneficial for accurate and interpretable detection of bipolar disorder. In the future, we plan to extend our method to a broader range of mental health analysis and mental illness detection tasks.

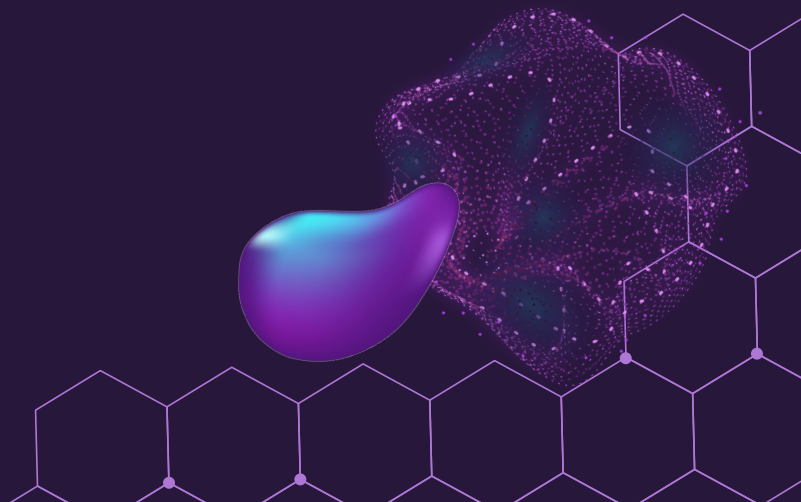
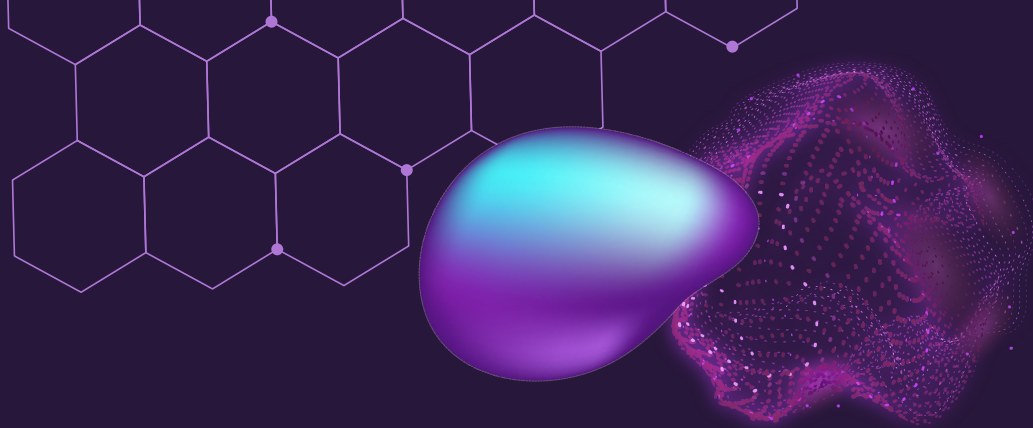
The background is a dark purple gradient. It features several abstract elements: a hexagonal grid pattern in the top left and bottom right corners, rendered in a lighter purple line art style; a large, smooth, glowing sphere with a blue-to-purple gradient in the upper right; a smaller, similar sphere in the lower right; and a complex, multi-layered structure of glowing purple points and lines in the center right, resembling a molecular or data structure.

04

**GET TO KNOW YOUR GROUP
AND START WORKING ON IT**

05

FOR NEXT TIME...





A 5 minutes presentation WITH SLIDES

On the Research Problem Assignment.

No presentation, no feedback before the
assignment' submission :)

